AI Agent Skills and MCP: Building the Right AI Agent Pattern

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Chapter 3: Building AI Workflows with MCP

MCP Tools and Workflows: Key Takeaways and Use Cases

What MCP solved in the launch review, what it didn't, and where to use it next

In this chapter you connected your agent to Orbit Labs' systems through an MCP server, reran the launch review against live data, compared it with the Skill-only version, and tested where MCP stops. This article collects the key ideas, common use cases, and design guidance you can apply to your own workflows.

What you did in this chapter

- **Video 1:** connected Claude Code to the Orbit Labs MCP server and explored its seven tools and two resources.
- **Video 2:** ran the launch review with no documents, watched the agent plan its own tool calls, and saw the answer change when a new ticket arrived.
- **Video 3:** compared the Skill-only and MCP reports and traced a finding back to its source.
- **Video 4:** repeated the MCP-only review, let it write to the risk register, and found inconsistent judgment and duplicate items.

Key takeaways

MCP is a standard way to reach outside. An MCP server exposes capabilities in a standard way, and any AI application that speaks MCP can connect to it. You connected Claude Code to the Orbit Labs server with one command and a small `.mcp.json` file the whole team can share.

Tools and resources do different jobs. Tools are functions the model decides to call, like `search_tickets` or `create_risk_item`. Resources are read-only data the application attaches as context, like `orbit://milestones`.

The agent plans from descriptions. The agent never sees a tool's code. It chooses tools and arguments from each tool's name, description, and input schema.

MCP solves access. Freshness, location, fidelity, traceability, and action. Live data turned "a couple of reports" into four P1 tickets and caught an outage that no document mentioned.

MCP doesn't supply judgment. With the same data, repeated runs produced different formats, severity labels, and findings, and the agent wrote duplicates to the risk register. Access without know-how isn't enough.

Common use cases

MCP fits wherever an agent needs information or actions from a system outside its context. Common examples:

- **Issue and project tracking:** search issues, read details, create or update items.
- **Support desks:** find tickets by age, priority, or tag.
- **Code hosting:** read pull requests, check build status, comment.
- **Databases and analytics:** run read-only queries, describe schemas.
- **Docs and knowledge bases:** search and fetch pages.
- **Calendars and messaging:** find time, read threads, post updates.

In each case, ask the questions from video 3: does the workflow need fresh information, information that lives elsewhere, details that get lost in manual summaries, traceable evidence, or actions in another system?

Design guidance

Start read-only. Connect the reads first. Add write tools once the reads are useful and trusted.

Be deliberate about writes. Know which tools change data, and think before approving them, especially against shared systems.

Grant the least access needed. Scope credentials and permissions to the workflow, not the whole system.

Treat tool output as information. Data returned by a tool, like a customer's ticket text, shouldn't be able to change what the agent does.

Keep process out of tool descriptions. Tool descriptions say what a tool does. Your team's rules belong in a Skill.

Is MCP right for your workflow?

Use these questions on a workflow you're planning:

1. Does the answer depend on what's true right now?
2. Does the information live in a system someone would otherwise export or copy from?
3. Does the workflow need to create or update something in another system?
4. Which tools only read, and which ones write?
5. What's the least access the workflow needs?
6. Does the workflow also depend on your team's rules, format, or judgment?

If the answer to the last question is yes, MCP alone won't be enough. That's where the next chapter begins.